

1. Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated

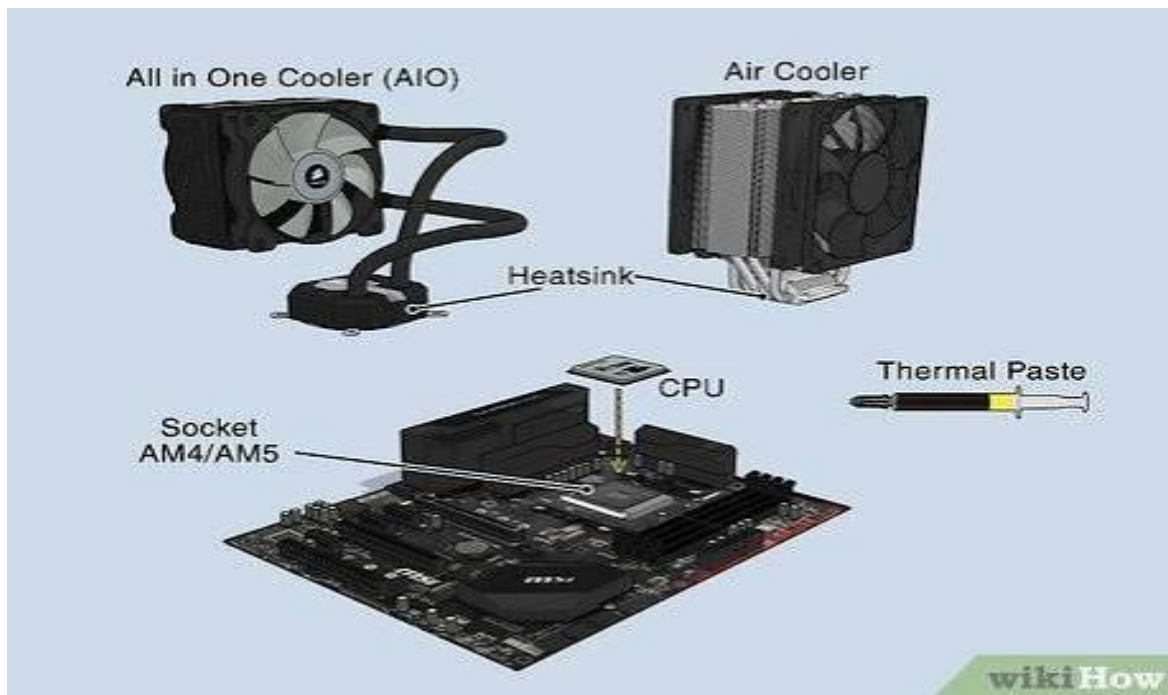
Brand: Intel

Model Number: Iris Xe Graphics

Type: Integrated

2.

Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.



1. Cooling Fan:

What it does: It works like a regular fan. It blows cool air over the heatsink to blow the hot air away from the computer.

2. Heatsink (Aluminum Fins):

Provides a large surface area to disperse heat into the air. The dense stack of metal fins absorbs heat transferred from the heat pipes.

3. Thermal Paste (Interface Material):

A thin layer of conductive paste applied between the CPU lid and the cooler baseplate. It fills microscopic air gaps to ensure smooth, direct heat transfer between the processor and the cooler.

3.Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Feature	Air Cooling	Liquid Cooling
Cooling Efficiency & Performance	Efficient for standard use and moderate gaming, but speeds up thermal throttling under extreme or prolonged workloads.	Liquid transfers heat away from the GPU die much faster than air, keeping GPU temperatures significantly lower under high workloads and heavy overclocking.
Cost & Budget	Standard air-cooled GPUs are cheaper to buy and require no additional expensive cooling hardware.	All-In-One (AIO) liquid-cooled GPUs or custom water loop blocks/radiators are significantly more expensive.
Maintenance & Longevity	Very Low Maintenance: Long-lasting with very few failure points. Only requires periodic cleaning with compressed air to clear dust from fan blades and heatsink fins.	Higher Maintenance: Radiator fans and fins gather dust quickly. Liquid pumps can wear out over time, and custom loops require fluid flushes, leak checks, and monitoring.

4. Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

My Choice:

Liquid Cooling I choose liquid cooling because playing heavy games and streaming IPL matches at the same time makes the GPU very hot. Liquid cooling keeps the system much cooler than regular fans, so the PC runs fast without slowing down. It is also very quiet, so no loud fan noise will ruin my live stream sound.